DRAFT – Updated Escapement Estimate Type Classification Guidance

NuSEDS Escapement Estimates Toolkit Working Group[1]

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2025

Canadian Technical Report of Fisheries and Aquatic Sciences to be assigned

Canadian Technical Report of Fisheries and Aquatic Sciences

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Introduction

DRAFT — Working document (do not cite)

The New Salmon Escapement Database System (NuSEDS) stores salmon spawner survey records, spawner abundance estimates, and linkages between them for Pacific Region salmon populations (Fisheries and Oceans Canada 2016). In NuSEDS, estimate Types 1–6 are used as a shorthand indicator of how interpretable an estimate is for downstream uses such as stock assessment and Wild Salmon Policy workflows.

In practice, Type assignment has varied across programs and areas because the historical table-only guidance does not fully specify how to treat incomplete coverage, uncertain methods, and other recurring real-world complications.

This update was prompted by inconsistent application across operational teams, which leads to inconsistent reporting in NuSEDS and increases the risk of downstream misinterpretation—especially near screening thresholds used in rapid status workflows (Pestal et al. 2023; Fisheries and Oceans Canada 2024).

Purpose and scope

The purpose of this technical report is to provide forward, implementation-ready guidance for assigning NuSEDS estimate Types 1–6 in a way that is reproducible and interpretable to downstream data users.

This report:

- documents the decision logic for assigning Types 1–6, including standardized qualifier codes that record why an estimate was conservatively downgraded;
- clarifies evidence expectations for common operational complications;
- identifies a minimal set of metadata elements needed for interpretable Type assignment, with an emphasis on Stream Escapement Narratives (**SENs**) used to support NuSEDS uploads and review; and
- provides canonical roll-ups and mapping guidance for enumeration and estimate methods (including legacy-to-canonical term reconciliation) to support consistent forward use in NuSEDS and clearer interpretation of legacy method terms (Appendix C).

This report does **not** re-estimate escapement, evaluate biological status, or replace program-specific expert judgement.

Hyatt (1997) intent and foundation

Hyatt (1997) developed a draft, hierarchical escapement classification key intended to be “unambiguous, easy to apply and reliable” for both field personnel and informed data users

(Hyatt 1997). Hyatt noted that the key should be repeatable among users given the same information, useful both **before** surveys (planning) and **after** surveys (reporting/quality control), and applicable retrospectively to historic estimates stored in the regional database (Hyatt 1997).

Hyatt framed classification around three linked dimensions: (i) properties of the survey and analytical methods used, (ii) statistical properties of the resulting estimate (units, accuracy, and precision), and (iii) the level of documentation supporting interpretation (Hyatt 1997). The NuSEDS estimate-type table is derived from this foundation. This report does **not** reproduce Hyatt's original draft key verbatim; instead, it operationalizes Hyatt's intent as a property-first decision sequence that can be applied consistently to contemporary workflows.

Estimate type summary (Types 1–6)

Table 1 summarizes Types 1–6 for orientation. Types summarize interpretability given method properties, uncertainty evidence, and documentation; they do not guarantee unbiasedness.

Table 1: High-level summary of estimate Types 1–6 for orientation.

Type	Label	Units	Interpretation
1	Absolute abundance (census / near-census)	Absolute abundance	A direct count (or near-census) with high confidence that missed fish are negligible or well constrained.
2	Absolute abundance estimate (qualified)	Absolute abundance (often with uncertainty)	A standardized estimation method producing absolute abundance with qualified accuracy and, where available, quantified precision.
3	Relative abundance index (high resolution)	Relative units (index; within-series)	A consistent, high-effort index that supports within-series comparisons; absolute abundance and uncertainty may not be fully quantified.
4	Relative abundance index (medium resolution)	Relative units (index; within-series)	A lower-effort index where comparability is more sensitive to timing, visibility, and coverage.
5	Relative abundance /	Relative units (uncertain)	A numeric estimate where method information, documentation, or key qualifiers are insufficient to

Type	Label	Units	Interpretation
	estimate (low evidence)		support stronger interpretation.
6	Presence / not detected	+ / -	A non-numeric record indicating adults present or not detected with reliable species identification.

Why an update is needed

In operational datasets, the same nominal estimate Type can be assigned to methodologically different estimates, and the historical table-only guidance does not encode several recurring qualifiers that affect interpretation. Key drivers include:

- **Modern method coverage:** hydroacoustic imaging sonar approaches (e.g., DIDSON/ARIS) are increasingly used, but are not always clearly represented in table-only guidance (Holmes et al. 2005).
- **Coverage/uptime and breach/bypass:** device outages, incomplete run-window coverage, and breach/bypass events can materially affect interpretability unless magnitude and correction methods are documented (See et al. 2021).
- **Timing, visibility, and effort standardization:** for survey-based indices, timing relative to run timing, observer efficiency, and visibility often govern comparability (Jones et al. 1998; Korman et al. 2002; Holt and Cox 2008).
- **Combined-method workflows:** multi-component estimates require explicit component recording to avoid concealing a weaker component (Parsons and Skalski 2010).
- **Downstream screening sensitivity:** estimate Types are used as data-quality filters in Wild Salmon Policy rapid status workflows; inconsistent assignment can create discontinuities near threshold cutoffs (Pestal et al. 2023; Fisheries and Oceans Canada 2024).

How this guidance should be used when uploading to NuSEDS

This update is intended to support consistent Type assignment during data entry and review by making key qualifiers explicit.

In practice:

- **Record methods explicitly** (enumeration and estimation). If methods are unknown, the guidance assigns a conservative Type 5 and records a method-unknown qualifier.

- **Record key qualifiers** that affect interpretation (run-window coverage, uptime, breach/bypass context, timing/visibility constraints, and any infilling/interpolation method).
- **Provide documentation evidence** sufficient for independent interpretation (e.g., field logs, methods/QA notes). Where evidence is missing, the guidance applies conservative downgrades.
- **Report quantitative uncertainty when available** (e.g., CV or SE) to support qualified precision for Type 2 where appropriate (e.g., AUC and mark–recapture uncertainty reporting) (English et al. 1992; Schwarz et al. 1993; Hilborn et al. 1999; Parken et al. 2003).

What this report provides

Relative to table-only interpretation, the updated guidance:

1. **Separates enumeration from estimation.** Enumeration methods (field data collection) are recorded distinctly from estimation methods (analysis and modelling), so eligibility rules and qualifiers are applied transparently.
2. **Defines method families.** The many enumeration and estimation methods observed in NuSEDS are grouped into a small set of method families to support consistent property-based checks (see the *Method families* section).
3. **Implements a property-first decision sequence.** In plain language: confirm record type (numeric vs presence/not detected), confirm methods are known, select a method family, apply method-family checks, and then apply final documentation/uncertainty gates (see the *Property-first decision key* section).
4. **Makes downgrade reasons explicit.** Conservative qualifiers (e.g., coverage/uptime, breach/bypass context, timing/visibility constraints, missing documentation, and quantified uncertainty when available) are recorded as explicit qualifier codes rather than as implicit judgement.
5. **Recommends minimal metadata for interpretability.** The guidance identifies a minimal set of metadata elements that must be captured for Type assignment to remain interpretable to downstream users (see the *Recommended metadata to record* section).

Classification framework and decision key

Design principles (Hyatt-aligned)

Hyatt (1997) intended estimate-type assignment to reflect (i) method properties, (ii) statistical properties, and (iii) documentation (Hyatt 1997). This report makes those dimensions explicit as decision gates, method-family checks, and final evidence requirements.

Practically, classification depends on **two method layers**:

- **Enumeration method** (field data collection)
- **Estimation method** (analysis/model that turns observations into an estimate)

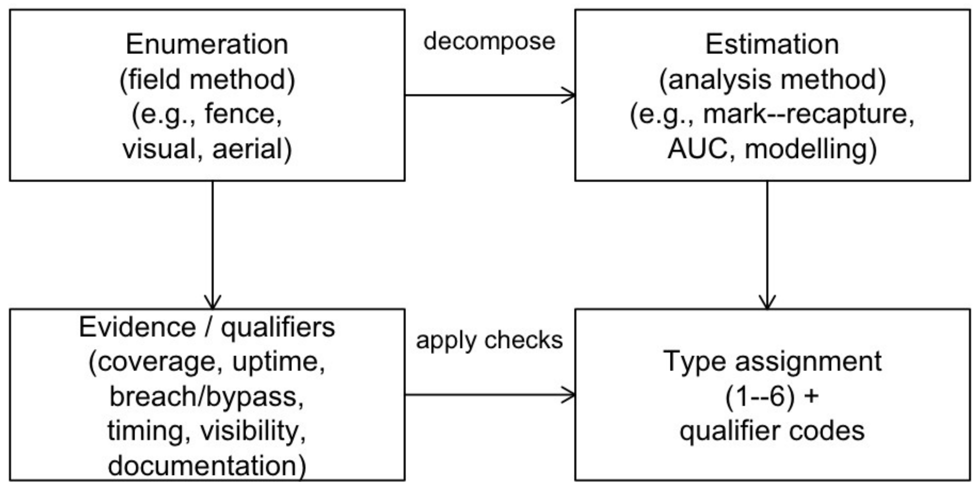

Figure 1. Conceptual decomposition of enumeration (field) methods and estimation (analysis) methods used to assign estimate Types and qualifiers.

Decision key design

The classification follows a property-first sequence:

1. **Data format gate**: non-numeric presence/not-detected is classified as Type 6.
2. **Method known gate**: if survey and analytical methods are not identified, the estimate is provisionally Type 5 and flagged as method-unknown.
3. **Method family selection**: a primary method family scopes the applicable questions.
4. **Method-family checks**: coverage, effort, visibility, timing, and related criteria drive conservative downgrades.
5. **Final checks**: documentation and uncertainty evidence are evaluated; where evidence is missing, conservative downgrades apply.

This structure reduces subjective table interpretation by turning common qualifiers into explicit questions and recorded qualifier codes.

Escapement Estimate Type Toolkit (informative)

An accompanying Escapement Estimate Type Toolkit (R Shiny application) applies the same decision logic described in this report and returns both (i) a **Final Type** (1–6) and (ii) qualifier codes explaining any conservative downgrades. The report remains the authoritative guidance; the toolkit is a practical companion implementation.

Within the toolkit, three intermediate labels are used for workflow transparency: **Provisional Type** (the type supported by the method-family path before final gates), **Max Type** (the upper bound imposed by triggered downgrade rules), and **Candidate Type** (the type implied after combining the enumeration and estimation steps before final documentation/precision checks). The report is concerned with the **Final Type** reported after all qualifier enforcement.

Guidance-to-implementation outlook

Implementation should align with the guidance and method-family checks in this report. Priorities are to:

- capture complete enumeration and estimation methods for each estimate;
- apply mandatory qualifiers where applicable; and
- track unresolved policy/specification items (e.g., mixed-method combination handling and ambiguity thresholds) through a controlled revision cycle.

Handling common operational complications

The guidance treats several recurring complications as explicit qualifiers rather than implicit judgement calls.

Breaches/bypass and infilling

Breach/bypass events at counting sites, and periods of missing observation due to outages or missed visits, can materially affect interpretability unless magnitude and correction methods are documented. The updated guidance treats breach/bypass risk and incomplete coverage as explicit downgrade triggers and records whether defensible infilling/interpolation was required (Holmes et al. 2005; Vélez-Espino et al. 2010; See et al. 2021). For example, run-window coverage below 95% of season days or device uptime below 95% are treated as explicit qualifiers, and any interpolation used to fill missing data should be recorded.

Survey timing relative to run timing

For survey-based indices, visit count alone is not sufficient: surveys must bracket the period when fish are present, and visibility constraints and observer effects can govern bias and comparability (Hill 1997; Jones et al. 1998; Korman et al. 2002; Holt and Cox 2008). The updated guidance includes timing and visibility checks within method families where these factors are primary drivers of interpretability.

Combined-method estimates

Combined-method workflows (e.g., system-wide sonar with tributary visual apportionment, or fences supplemented during breach periods) should be recorded as explicit components. The updated guidance applies a conservative rule: where multiple components contribute to the final estimate, the assigned Type should not exceed that implied by the weakest component unless a documented integration method supports a higher classification (Parsons and Skalski 2010).

Calibration and historical revisions

Where historical values have been recalibrated or revised, the updated guidance treats calibration as an analysis layer that should be captured in metadata (calibration source, diagnostics, and revision history) rather than as a new estimate Type. This supports interpretation of time-series values by downstream users, but this report does **not** prescribe automatic reclassification of existing records or a downstream assessment rule for calibrated series.

Quantitative uncertainty (CV/SE)

Hyatt (1997) distinguished higher-quality absolute-abundance estimates in part by qualified precision (variance evidence) (Hyatt 1997). Where available, reporting quantitative uncertainty (e.g., CV or SE) improves interpretability for downstream use. For example, AUC and peak-count approaches have established approaches for incorporating uncertainty and evaluating sensitivity to survey timing and frequency (English et al. 1992; Hill 1997; Hilborn et al. 1999; Parken et al. 2003; Millar et al. 2012). The updated guidance includes a final precision/accuracy check and recommends capturing quantitative uncertainty when available.

Recommended metadata elements for NuSEDS

Most of the guidance can be expressed using existing NuSEDS fields, but several high-value metadata elements are needed to make estimate Types interpretable and reproducible for downstream users.

The table below lists guidance-level metadata elements and the typical qualifier codes they support; detailed software rollout and database-change logistics are treated as implementation work outside the scope of this technical report.

Table 2: Recommended metadata elements for NuSEDS uploads to support transparent Type 1–6 assignment and interpretation.

Metadata element	Why it matters for interpretation	Typical qualifier code(s)	Priority for upload
Enumeration	Scopes which property checks apply.	METHOD_UNKNOW	Require

Metadata element	Why it matters for interpretation	Typical qualifier code(s)	Priority for upload
method (field)		N (if missing)	d
Estimation method (analysis)	Distinguishes analysis pathways (e.g., mark–recapture, AUC, modelling).	METHOD_UNKNOWN (if missing)	Required
Primary method family (FS/V/A/S/T/R/P/M)	Controls eligibility rules and the set of required qualifiers.	METHOD_UNKNOWN (if missing)	Required
Run-window coverage evidence	Explains conservative downgrades due to missed start/end of run.	RUN_COVERAGE	Recommended
Device/site uptime evidence	Explains conservative downgrades due to outages/unobserved intervals.	UPTIME	Recommended
Breach/bypass context	Explains conservative downgrades where missed fish are plausible.	BREACH_BYPASS	Recommended
Infilling/ interpolation method	Distinguishes defensible correction from undocumented gap-filling.	INFILL_METHOD	Recommended
Survey timing relative to run timing	A core driver of bias and comparability for survey indices.	TIMING	Recommended
Visibility/ conditions	Affects detectability and comparability (especially for visual methods).	VISIBILITY / ENV_COND	Recommended
Documentation evidence	Determines whether a downstream user can independently interpret the estimate.	DOC	Required
Quantitative uncertainty (CV/SE)	Supports qualified precision for Type 2 and more transparent uncertainty.	PRECISION_ACCURACY	Recommended
Combined-method components	Prevents concealment of a weaker component within an overall estimate.	Component-level qualifiers + integration rule	Recommended
Calibration source + diagnostics	Supports interpretation of recalibrated time series.	Calibration diagnostics / revision	Recommended

Metadata element	Why it matters for interpretation	Typical qualifier code(s)	Priority for upload
		metadata	
Revision history (what changed/when)	Supports transparency when historical values have been revised.	Revision metadata / audit trail	Recommended

Method families

Families are listed by the short code used throughout the report and toolkit, rather than by attainable Type.

Table 3: Method families encoded in the property-first guidance and the best attainable Type before conservative downgrades.

Code	Method family	Best attainable type
FS	Fixed site census / near-census (manual or validated counter tally)	1
V	Visual ground or snorkel count	2
A	Aerial survey count	3
S	Hydroacoustic sonar count (modelled)	2
T	Trap model (non-spanning)	2
R	Redd survey	3
P	Electrofishing CPUE index	3
M	Mark-recapture program	2

Type 1 retains an interpretive meaning—**absolute abundance from a census / near-census workflow**—rather than serving as a raw method label. In practice, only constrained fixed-site tallies can default to that upper bound under the current guidance.

Analysis labels such as AUC and peak-count do not replace the underlying survey family. Ground/snorkel series analysed with AUC can still support Type 2 under the V family, whereas aerial series analysed with AUC or peak-count remain capped by the A family ceiling (Type 3) unless a future validated exception is explicitly adopted.

Default redd-count pathways are conservatively capped at Type 3 in this update. Any higher-confidence redd-derived absolute-abundance estimate should be treated as an explicit calibrated/model-based exception rather than as the default redd family interpretation.

DFO Salmon Data Standards and Controlled Vocabularies linkages for methods and downgrade criteria

To support consistent interpretation (including cross-program interoperability), these NuSEDS elements can be linked to the DFO Salmon Data Standards and Controlled Vocabularies (DFO Salmon Ontology) term browser and concept schemes:

- https://w3id.org/gcdfo/salmon

Table 4: Ontology concept-scheme linkages for NuSEDS enumeration methods, estimate methods, and downgrade criteria.

NuSEDS element	Ontology scheme term	Ontology IRI	WIDOCO term page
ENUMERATION_METHODS	gcdfo:EnumerationMethodScheme	https://w3id.org/gcdfo/salmon#EnumerationMethodScheme	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#EnumerationMethodScheme
ESTIMATE_METHOD	gcdfo:EstimateMethodScheme	https://w3id.org/gcdfo/salmon#EstimateMethodScheme	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#EstimateMethodScheme
DOWNGRADE_CRITERIA	gcdfo:DowngradeCriteriaScheme	https://w3id.org/gcdfo/salmon#DowngradeCriteriaScheme	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#DowngradeCriteriaScheme

Detailed term-level crosswalks are provided in Appendix B (downgrade criteria) and Appendix C (enumeration and estimate methods). Appendix C also includes compact Sankey-style summaries to visualize how legacy NuSEDS method terms converge to the canonical parent terms used by the guidance. Equivalent lookup helpers are also exposed in the metasalmon package (nuseds_enumeration_method_crosswalk() and nuseds_estimate_method_crosswalk()) (DFO Pacific Science 2026).

Legacy values are not dropped; where a legacy term does not cleanly align to a current method family, it can be represented as inherited contextual guidance via ontology metadata (for example via rdfs:comment) while remaining traceable in source records.

NuSEDS data dictionary alignment (selected fields)

The NuSEDS data dictionary defines the database fields used to store enumeration methods, estimation methods, estimate classification (Types 1–6), and supporting metadata (e.g., inspections/effort and timing fields) (Fisheries and Oceans Canada 2025). The updated guidance is designed to be expressible using existing NuSEDS fields where possible, and to clearly identify gaps where additional metadata would improve interpretation and reproducibility.

Table 5: NuSEDS fields relevant to estimate-type classification and how they relate to the updated guidance (see the NuSEDS data dictionary in docs/context/).

	Field Name	Role in updated guidance
3	ESTIMATE_CLASSIFICATION	Stores Type 1–6 estimate classification (including legacy non-Type labels where present)
2	ENUMERATION_METHODS	Primary field method (enumeration) used to scope method-family checks
4	ESTIMATE_METHOD	Primary analysis method (estimation), including combined or calibrated special cases
7	NO_INSPECTIONS_USED	Supports effort/visit thresholds (VISITS and related downgrades)
6	INDEX_YN	Flags index (partial-coverage) estimates (relative-abundance context)
1	ACCURACY	Legacy qualitative uncertainty field; not a substitute for quantified metadata
8	PRECISION	Legacy qualitative uncertainty field; not a substitute for quantified metadata
9	RELIABILITY	Legacy/import field (historical); not consistently present
5	ESTIMATE_STAGE	QA/workflow stage (preliminary/near final/final); not a type determinant

Applying the guidance in practice

Next we summarize the substantive updates to the Type 1–6 guidance and how it should be applied during NuSEDS data entry and review.

What is new in this guidance

Compared to table-only interpretation, the updated guidance:

- Uses a property-first decision sequence (rather than a table lookup).
- Separates enumeration methods (field) from estimation methods (analysis).
- Applies explicit qualifier codes for conservative downgrades (coverage, timing, visibility, documentation, and method-known gaps).
- Treats combined-method estimates as component-based workflows so weaker components are not concealed.
- Encourages recording quantitative uncertainty (CV/SE) where available.

Legacy method vocabulary context

NuSEDS stores historical method terminology across multiple legacy labels. To support implementation planning, we reviewed method distributions in both ENUMERATION_METHODS and ESTIMATE_METHOD and retained the most frequent terms for interpretability, with longer-tail terms preserved through controlled mapping in Appendix C.

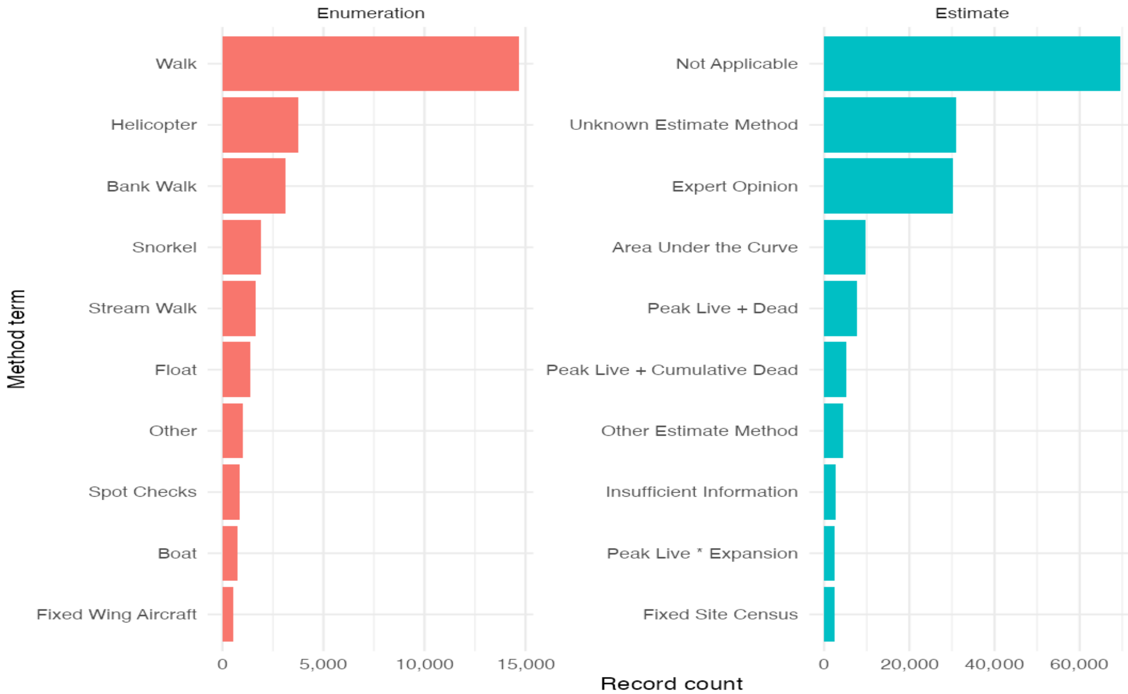

Figure 2. Top 10 method terms in NuSEDS SEN method fields (enumeration and estimate), based on the all-area SEN method extract assembled for this report (analysis years 1990–2025).

These frequencies should be read as a dated implementation snapshot rather than as a stable inventory of all future NuSEDS values. Appendix D summarizes the source and processing notes used for this exploratory method-term review.

How to apply the guidance moving forward

For most users, the simplest way to apply this guidance is through the **Escapement Estimate Type Toolkit** (R Shiny application). A current ShinyApps.io deployment is available at:

- https://dfo-pac-sci-dev.shinyapps.io/smn-escapement-toolkit/

The toolkit guides users through the same decision sequence described in this report and produces (i) a **Final Type** (1–6) and (ii) explicit qualifier/downgrade codes. The report remains the authoritative guidance; the toolkit is the operational companion. For operational entry, users should record the toolkit **Final Type** as the estimate-type classification result, store qualifier codes in DOWNGRADE_CRITERIA, and use ESTIMATE_TYPE_CLASS_JUSTIFICATION plus ANALYSIS_EXPLANATION to capture the rationale and any combined/calibrated method detail.

Max Type, Provisional Type, and Candidate Type are toolkit-internal QA trace values and should not be treated as the official uploaded classification.

Authoritative decision-key logic and change requests

The authoritative decision-key logic is maintained as a version-controlled YAML file in the toolkit repository:

- https://github.com/dfo-pacific-science/smn-escapement-estimates-toolkit/blob/main/matrix_key/structured_dichotomous_key.yaml

Requests to update the logic (new method terms, revised thresholds, clarified combination rules, etc.) should be submitted as a GitHub issue in that repository (preferred) or emailed to **FADSDataStewardship-GestiondesdonneesSFDA@dfo-mpo.gc.ca**.

What must be recorded

Uploads should include enough method and evidence context that downstream users can interpret both the assigned Type and any conservative qualifiers. For teams that still populate legacy accuracy/precision labels alongside estimate Type, Appendix C.3 documents the compact lookup currently used in the toolkit. In practice, this means recording:

- the primary enumeration and estimation methods;
- key qualifiers (coverage, uptime, breach/bypass, timing, visibility, and any infilling/interpolation context);
- documentation evidence sufficient for independent interpretation; and
- quantitative uncertainty where available.

A compact, single-table rule summary is provided in Appendix A.

Implications, limitations, and governance

Intended use (NuSEDS data entry and review)

This guidance supports consistent estimate-type assignment during data entry and review by:

- making method and evidence expectations explicit (not implicit in table lookups);
- recording conservative qualifier codes that explain why a candidate Type is not supported; and
- improving interpretability by separating enumeration (field) methods from estimation (analysis) methods.

Estimate Types are not a binary “usable/unusable” label. Lower Types may still be fit for specific purposes (e.g., presence/absence, qualitative local context, or within-program indices), but they are not interchangeable with census-like or fully qualified absolute-abundance estimates.

Implications for downstream assessments

Estimate Types are often used as screening filters in downstream workflows (including WSP rapid status assessments) (Fisheries and Oceans Canada 2005, 2024; Pestal et al. 2023). When Type assignment is inconsistent, threshold-based decisions can become sensitive to classification noise, particularly near cutoffs (e.g., around the Type 4 boundary). More broadly, measurement error in spawner abundance can distort stock–recruit inference and downstream decisions, reinforcing the value of recording uncertainty and detectability-related metadata where available (Walters and Ludwig 1981).

Because qualifier codes are explicit (e.g., missing method identification, timing/coverage gaps, insufficient documentation), conservative classifications become interpretable and metadata improvements can be targeted.

Recommended metadata additions (schema) and implementation status

A central finding of this update is that the Type table alone cannot communicate the information required for consistent interpretation of real-world estimates. Several recurring qualifiers are better represented as explicit metadata elements rather than as new public estimate Types.

Based on working-group implementation notes, near-term rollout is best framed as **SEN-first** with controlled vocabularies. Here, **SEN-first** means prioritizing the fields that can already be captured in Stream Escapement Narratives (SENs) and their associated upload templates before extending site-information-layer or CU-context schema.

SEN fields prioritized for implementation

The following SEN fields are the clearest priorities for aligning NuSEDS uploads with the updated guidance:

- ESTIMATE_TYPE_CLASS_JUSTIFICATION: basis for the Type assignment.
- DOWNGRADE_CRITERIA: which downgrade trigger(s) were applied.
- UNCERTAINTY_VALUE_TYPE: uncertainty metric class (e.g., CV, SE, CI, SD).
- UNCERTAINTY_VALUE: corresponding numeric uncertainty value.
- ANALYSIS_EXPLANATION: analysis details, including modified or combined methods.

Together, these fields operationalize core qualifiers in the decision key: method clarity, downgrade rationale, and uncertainty transparency. In operational use, the uploaded classification should be the toolkit **Final Type**; Max Type, Provisional Type, and Candidate Type remain internal decision-trace fields for QA rather than official classification values. This report identifies what should be captured for interpretation; analyst-facing availability in downstream exports is an implementation question outside the scope of this report.

Supporting lookup tables and naming updates

Supporting structures already partway through rollout include:

- DOWNGRADE_CRITERIA_BASE_TABLE: controlled values for downgrade criteria.
- UNCERTAINTY_VALUE_TYPE_BASE_TABLE: controlled values/definitions for uncertainty metric types.
- DOWNGRADE_CRITERIA_IMPACT renamed to DOWNGRADE_CRITERIA_COMMENT to support broader explanatory context.

Controlled values reduce ambiguity and help keep entries consistent across areas and programs.

Items still under discussion

Several metadata items remain valuable but require further specification/governance:

- site-information-layer additions are less mature than SEN additions and require further prioritization;
- ENHANCEMENT_RANK, POP_CATEGORY, and CUMatchVerifiedYN are potential follow-on site/CU context fields for WSP-oriented interpretation; and
- a dedicated WSP rapid-status flag is **not** recommended in this report because inclusion/exclusion of systems can change over time and would require active governance and maintenance.

For this report, the implementation priority is therefore: finalize SEN metadata first, keep site/CU context additions explicit as backlog, and maintain a transparent mapping from stored fields to decision-key qualifiers.

Escapement Estimate Type Toolkit: unresolved policy/specification gaps

The Escapement Estimate Type Toolkit (R Shiny application) surfaces policy/specification gaps that can still affect cross-program consistency if left implicit. The most material items are:

- **Mixed-method estimates:** explicit combination policy is needed (e.g., conservative minimum-component rule vs. weighted integration with evidence requirements).
- **Ambiguous threshold language:** terms such as “negligible”, “adequate”, “fair”, and “defensible” still need agreed quantitative thresholds.
- **Breach severity interpretation:** clearer rules are needed for how breach magnitude/timing (e.g., peak vs early/late-season) should influence downgrades.
- **Calibration diagnostics standards:** calibrated time-series pathways need explicit minimum diagnostics/validation expectations.
- **Documentation standardization:** documentation labels are functionally similar but still inconsistently phrased across method pathways.

- **Type-5 pathway clarity:** low-resolution relative-estimate pathways should remain explicit to avoid ad hoc interpretation.

This backlog reflects remaining work for the Regional Escapement Task Team. Perfect cross-program consistency and zero ambiguity may not be feasible given the diversity of operational contexts; the goal is to keep the most consequential rules explicit, versioned, and reviewable.

Limitations

This guidance summarizes interpretability given recorded methods and evidence. It does not correct biased inputs, replace program expertise, or guarantee biological accuracy. The most common practical limitation is incomplete metadata: when critical qualifiers are not recorded, the guidance applies conservative classification (often Type 5) to avoid overstating interpretability.

Governance and change control

Because estimate Types are used as screening tools and can influence downstream inference, changes to the guidance, method families, thresholds, or qualifier rules should be versioned and reviewed. Updates should be accompanied by worked examples and regression/path tests in the Escapement Estimate Type Toolkit so behavior changes are deliberate and transparent.

Conclusion

This report retains the familiar Type 1–6 structure but replaces table-only interpretation with explicit, auditable decision logic grounded in Hyatt (1997): method properties, statistical properties, and documentation (Hyatt 1997).

For NuSEDS users, the practical change is simple: record methods and key qualifiers at upload time so Types remain interpretable downstream. Run-window coverage, uptime, breach/bypass context, timing/visibility constraints, documentation evidence, and (where available) quantitative uncertainty (CV/SE) affect how an estimate can be used.

For downstream analysts, the main benefit is transparency: conservative downgrades are accompanied by explicit qualifier codes, making it clear why a stronger Type is not supported and which metadata gaps matter most.

Next steps

- Finalize a minimal, public-facing metadata schema for NuSEDS that supports the guidance (including quantitative uncertainty fields where available).
- Maintain a small set of worked examples (anonymized as needed) to demonstrate expected behavior on representative cases.

References

Appendix

Appendix A: Classification rule summary

Table 6: Single-table summary of the NuSEDS estimate-type classification rules used for forward implementation.

Rule stage	Decision / trigger	Classification effect	Primary qualifier code(s)
1. Data format gate	Record is non-numeric (presence or not-detected)	Assign Type 6	(none)
2. Method known gate	Enumeration and/or estimation method not identified	Assign Provisional Type 5	METHOD_UNKNOWN
3. Method family assignment	Primary method family selected (FS, V, A, S, T, R, P, M)	Select applicable rule path	(path selection step)
4. Family-specific checks (counting systems)	Coverage, uptime, bypass/breach, cross-section, and related counting-system checks fail or are incomplete	Apply conservative downgrades within selected path	RUN_COVERAGE, UPTIME, BREACH_BYPASS, XSEC_COVERAGE, DEVICE_CONFIG
5. Family-specific checks (survey indices)	Visit effort, timing relative to run, visibility, reach coverage, or related survey checks fail or are incomplete	Apply conservative downgrades within selected path	VISITS, TIMING, VISIBILITY, REACH_COVERAGE, EFF_STD, DETECTABILITY
6. Combined-method integration rule	Multiple components contribute to final estimate	Final Type cannot exceed weakest component unless integration method supports stronger classification	INFILL_METHOD (as relevant), component-level qualifiers
7. Documentation gate	Evidence package insufficient for independent interpretation	Apply conservative downgrade	DOC

Rule stage	Decision / trigger	Classification effect	Primary qualifier code(s)
8. Precision/ accuracy gate	Candidate Type requires uncertainty/accuracy evidence that is missing or weak	Apply conservative downgrade	PRECISION_ACCURACY
9. Final assignment	All gates evaluated	Return Final Type plus explicit qualifier codes	All triggered qualifiers retained as audit trail

Appendix B: Downgrade criteria codes

The qualifier-code table is formatted for high-density reference use. In PDF output it is rendered on a landscape page to improve line wrapping and scanability; HTML and DOCX outputs retain standard page flow.

Table 7: Downgrade criteria codes used by the classification key, with linkages to published DFO Salmon Ontology terms.

Code	Description	Ontology term	Ontology IRI
RUN_COVERAGE	Run window coverage gaps (e.g., <95% of season days).	gcdfo:RunCoverage	https://w3id.org/gcdfo/salmon#RunCoverage
VISITS	Too few site visits for the claimed precision tier.	gcdfo:NumberOfVisits	https://w3id.org/gcdfo/salmon#NumberOfVisits
BREACH_BYPASS	Bypass or breach risk at a counting site.	gcdfo:BreachBypass	https://w3id.org/gcdfo/salmon#BreachBypass
UPTIME	Device outages or unobserved intervals (e.g., uptime <95%).	gcdfo:Uptime	https://w3id.org/gcdfo/salmon#Uptime
REVIEW_QA	Insufficient review of automated events.	gcdfo:ReviewQA	https://w3id.org/gcdfo/salmon#ReviewQA
VISIBILITY	Poor visibility or lighting limits detection.	gcdfo:Visibility	https://w3id.org/gcdfo/salmon#Visibility
REACH_COVERAGE	Intended reaches or segments not surveyed.	gcdfo:ReachCoverage	https://w3id.org/gcdfo/salmon#ReachCoverage
TIMING	Visits not aligned with migration	gcdfo:Timing	https://w3id.org/gcdfo/

Code	Description	Ontology term	Ontology IRI
	peak or plateau.		salmon#Timing
EFF_STD	Inconsistent effort standardization.	gcdfo:EffortStandardization	https://w3id.org/gcdfo/salmon#EffortStandardization
XSEC_COVERAGE	Partial cross-section coverage.	gcdfo:CrossSectionCoverage	https://w3id.org/gcdfo/salmon#CrossSectionCoverage
CLASSIFICATION	Species or target classification error.	gcdfo:Classification	https://w3id.org/gcdfo/salmon#Classification
TRAP_EFF	Unmeasured or uncertain trap efficiency.	gcdfo:TrapEfficiency	https://w3id.org/gcdfo/salmon#TrapEfficiency
DETECTABILITY	Redd or carcass detectability bias.	gcdfo:Detectability	https://w3id.org/gcdfo/salmon#Detectability
DEVICE_CONFIG	Device settings not documented or unstable.	gcdfo:DeviceConfiguration	https://w3id.org/gcdfo/salmon#DeviceConfiguration
ENV_COND	Environmental conditions bias detection.	gcdfo:EnvironmentalConditions	https://w3id.org/gcdfo/salmon#EnvironmentalConditions
MR_ASSUMP	Mark-recapture assumption violations.	gcdfo:MarkRecaptureAssumptions	https://w3id.org/gcdfo/salmon#MarkRecaptureAssumptions
INFILL_METHOD	Interpolation or infilling used; defensibility must be documented.	gcdfo:InfillMethod	https://w3id.org/gcdfo/salmon#InfillMethod
DOC	Documentation or evidence missing.	gcdfo:Documentation	https://w3id.org/gcdfo/salmon#Documentation
PRECISION_ACCURACY	Precision or accuracy weaker than candidate Type.	gcdfo:PrecisionAccuracy	https://w3id.org/gcdfo/salmon#PrecisionAccuracy
METHOD_UNKNOWN	Survey or analysis method not identified.	gcdfo:MethodUnknown	https://w3id.org/gcdfo/salmon#MethodUnknown

Appendix C: NuSEDS data dictionary crosswalk

This appendix documents how the updated guidance relates to NuSEDS fields and controlled vocabularies (see docs/context/Data_Dictionary_NuSEDS_EN.csv).

C.1 Enumeration method values (ENUMERATION_METHODS)

Table 8: Suggested mapping from NuSEDS enumeration method values to the updated key method families (FS, V, A, S, T, R, P, M), with ontology term and WIDOCO linkages.

NuSEDS ENUMERATION_METHODS value	Suggested method family	Ontology term	WIDOCO term page	Notes
Bank Walk	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	
Based on Angling Catch	P	gcdfo:EnumerationMethod	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#EnumerationMethod	Catch-based index; treat as a CPUE-style index unless more detail is provided; ontology linkage is provisional at scheme level.
Biologist/ Working Group	unknown	NA	NA	Not a method; treat as method-unknown unless a specific field/analysis method is documented elsewhere.
Boat	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	
Broodstock Removal	FS	gcdfo:FixedSiteCensusManual	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#FixedSiteCensusManual	
Dead Pitch	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	Carcass-based visual surveys; often paired with peak/cumulative dead estimation methods.

NuSEDS ENUMERATION_METHODS value	Suggested method family	Ontology term	WIDOCO term page	Notes
Electronic Counters	FS	gcdfo:FixedSiteCensusElectronic	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#FixedSiteCensusElectronic	
Electroshocking	P	gcdfo:ElectrofishingCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ElectrofishingCount	
Enumeration by Hatchery	FS	gcdfo:FixedSiteCensusManual	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#FixedSiteCensusManual	
Fence	FS	gcdfo:FixedSiteCensusManual	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#FixedSiteCensusManual	
Fixed Wing Aircraft	A	gcdfo:AerialSurveyCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#AerialSurveyCount	
Float	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	
Helicopter	A	gcdfo:AerialSurveyCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#AerialSurveyCount	

NuSEDS ENUMERATION_METHODS value	Suggested method family	Ontology term	WIDOCO term page	Notes
Hydroacoustic Station	S	gcdfo:HydroacousticSonarCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#HydroacousticSonarCount	
Other	unknown	NA	NA	Treat as method-unknown unless a specific field/analysis method is documented elsewhere.
Peak Live and Dead Count	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	Value is analysis-like; prefer capturing peak/cumulative variants under ESTIMATE_METHOD.
Redd Counts	R	gcdfo:ReddCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ReddCount	
Snorkel	V	gcdfo:VisualSnorkelCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualSnorkelCount	
Spot Checks	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	
Stream Walk	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	
Strip Counts	V	gcdfo:VisualGr	https://dfo-pacific-science.github.io/dfo-	

NuSEDS ENUMERATION_METHODS value	Suggested method family	Ontology term	WIDOCO term page	Notes
		oundCount	salmon-ontology/#VisualGroundCount	
Tag Recovery	M	gcdfo:MarkRecaptureFieldProgram	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#MarkRecaptureFieldProgram	
Trap	T	gcdfo:TrapCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#TrapCount	If trap is non-spanning or efficiency-corrected, use T; fully constraining traps may behave more like fixed-site counting.
Walk	V	gcdfo:VisualGroundCount	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#VisualGroundCount	

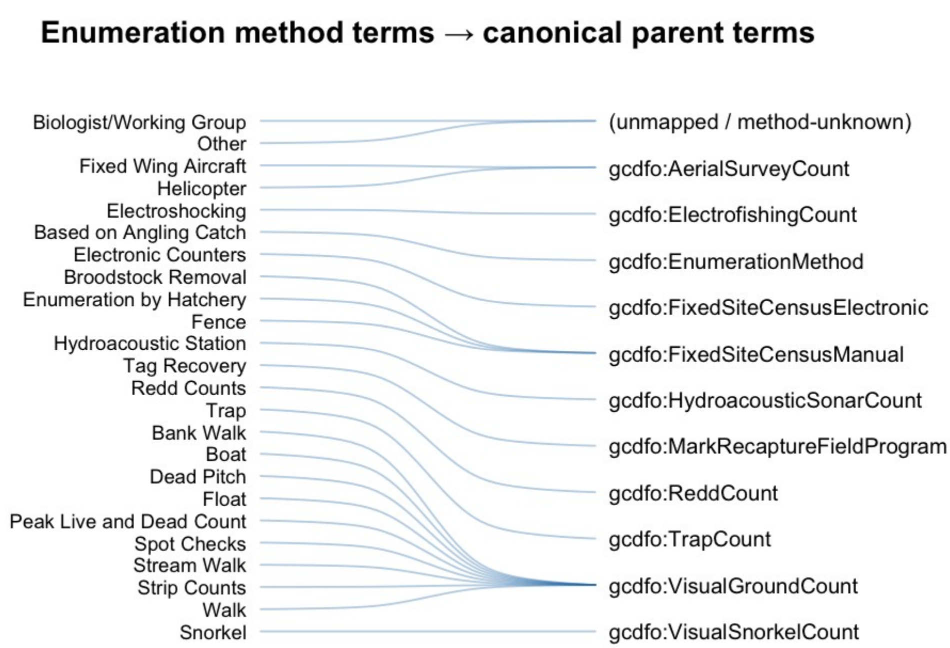

Figure 3. Sankey-style summary of how legacy NuSEDS enumeration-method terms converge to canonical parent terms used by the updated guidance.

C.2 Estimate method values (ESTIMATE_METHOD)

Table 9: Suggested interpretation of NuSEDS estimate method values under the updated guidance, with ontology term and WIDOCO linkages. Where 'depends' appears, additional method metadata are required to apply the key cleanly.

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Cumulative CPUE	CPUE index	P	gcdfo:EstimateMethod	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#EstimateMethod	No specific CPUE estimate concept is currently defined in this scheme; linked at EstimateMethod scheme level.
Fixed Site	Fixed-station tally / counter mode (often stored as estimate	FS	gcdfo:FixedStati	https://dfo-pacific-science.github.io/dfo-salmon-	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Census	method)		onTally	ontology/#FixedStationTally	
Cumulative New	Method unknown/administrative label	unknown	gcdfo:PeakCountAnalysis	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#PeakCountAnalysis	Legacy dictionary label with no method definition in the source description; mapped to PeakCountAnalysis pending local confirmation.
Peak Live + Dead	Survey-series estimation (AUC/peak variants; platform-specific ceiling depends on underlying	depends	gcdfo:PeakCountAnalys	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#PeakCount	Peak-count summaries inherit the underlying survey family. Aerial-derived peak summaries remain capped by the A-family ceiling unless a future explicit exception is adopted.

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
	survey family)		is	Analysis	
Area Under the Curve	Survey-series estimation (AUC/peak variants; platform-specific ceiling depends on underlying survey family)	depends	gcdfo:AreaUnderTheCurve	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#AreaUnderTheCurve	Treat AUC as an analysis label applied after survey-family selection: ground/snorkel series follow the V-family pathway, while aerial series remain capped by the A-family ceiling (Type 3) under the current conservative guidance.
Mark & Recaptur	Mark-recapture estimation	M	gcdfo:MarkRecaptureAnal	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#MarkRecaptureAnalys	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
e: Jolly-Seber			ysis	[is](is)	
Addition/Subtraction	Math/ expansion operations (depends on base method)	depends	gcdfo:ExpansionMathematicalOperations	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ExpansionMathematicalOperations	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Multiplication / Division	Math/ expansion operations (depends on base method)	depends	gcdfo: ExpansionMathematicalOperations	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ExpansionMathematicalOperations	
Mark & Reca	Mark-recapture estimation	M	gcdfo: MarkRecaptu	https://dfo-pacific-science.github.io/dfo-salmon-ontology/	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
pture: Petersen			reAnalysis	[#MarkRecaptureAnalysis]()	
ReddCount	Redd-based estimation (default conservative survey/index pathway unless separately calibrated/modelled)	R	gcdfo:ReddExpansionAnalysis	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ReddExpansionAnalysis	Default redd-count interpretation is conservatively capped at Type 3. Any higher-confidence redd-derived absolute-abundance estimate should be handled only as an explicit calibrated/model-based exception with supporting validation and documentation.
Peak	Survey-series estimation	de	gcdfo:	[https://dfo-pacific-]()	Peak-count summaries inherit the underlying survey family. Aerial-derived peak summaries

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Live + Cummulative Dead	(AUC/peak variants; platform-specific ceiling depends on underlying survey family)	pends	PeakCountAnalysis	science.github.io/dfo-salmon-ontology/#PeakCountAnalysis	remain capped by the A-family ceiling unless a future explicit exception is adopted.
Lake Expansion	Math/ expansion operations (depends on base method)	depends	gcdfo: ExpansionMathema	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ExpansionMathematic	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
			ticalOperations	[alOperations]()	
Insufficient Information	Method unknown/administrative label	unknown	NA	NA	
Peak Liv	Survey-series expansion (inherits the	dep	gcdfo: Exp	[https://dfo-pacific-science.gith]()	Expansion does not replace the underlying survey family; record the base method and any

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
e * Expansion	underlying survey family and any weaker component)	ends	ansionMathematicalOperations	[ub.io/dfo-salmon-ontology/#ExpansionMathematicalOperations]()	weaker component explicitly.
Unknown Estimat	Method unknown/administrative label	unknown	NA	NA	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
eMethod					
Not Applicable	Method unknown/administrative label	unknown	NA	NA	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Expert Opinion	Method unknown/administrative label	unknown	NA	NA	
Mark & Recapture: Bayesia	Mark-recapture estimation	M	gcdfo:MarkRecaptureAnalysis	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#MarkRecaptureAnalysis	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
n					
Other Estimate Method	Method unknown/administrative label	unknown	NA	NA	
Sonar-DIDSO	Hydroacoustic modelling pipeline	S	gcdfo: Hydroacousti	https://dfo-pacific-science.github.io/dfo-salmon-ontology/	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
N			cModelling	#HydroacousticModelling	
Calibrated Time Series	Calibrated time series (requires calibration source + diagnostics)	depends	gcdfo: CalibratedTimeSeries	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#CalibratedTimeSeries	For interpretability, record calibration source years, diagnostics, and revision history.
Resistivity	Fixed-station tally / counter mode (often stored as	FS	gcdfo: FixedS	https://dfo-pacific-science.github.io/dfo-	Treat as a fixed-station tally when the installation functions as the primary constrained-opening counter. Record species/event attribution, validation,

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Counter	estimate method)		tationTally	salmon-ontology/#FixedStationTally	detection-correction, and bypass/coverage/QA context in the supporting explanation fields; resistivity remains in the FS pathway under the current guidance.
Sonar-ARIS	Hydroacoustic modelling pipeline	S	gcdfo:HydroacousticModelling	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#HydroacousticModelling	
Mark & Re	Mark-recapture estimation	M	gcdfo:MarkReca	https://dfo-pacific-science.github.io/dfo-salmon-	

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
capture: Open Model			ptureAnalysis	ontology/#MarkRecaptureAnalysis	
Combined Methods	Combined-method workflow (requires explicit component listing)	depends	gcdfo: EstimateMethod	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#EstimateMethod	Decompose into components (e.g., sonar + visual apportionment; fence + visual during breach) and apply conservative classification.

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
Video Counter	Fixed-station tally / counter mode (often stored as estimate method)	FS	gcdfo:FixedStationTally	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#FixedStationTally	Treat as a fixed-station tally within FS; ensure QA review rate, attribution logic (if used), and uptime/coverage metadata are captured.
(Peak Live+Cum Dead) *E	Survey-series expansion (inherits the underlying survey family and any weaker component)	depends	gcdfo:ExpansionMathematicalOpera	https://dfo-pacific-science.github.io/dfo-salmon-ontology/#ExpansionMathematicalOperations	Expansion does not replace the underlying survey family; record the base method and any weaker component explicitly.

NuSEDS ESTIMATE_METHOD value	Guidance interpretation	Suggested method family	Ontology term	WIDOCO term page	Notes
xpansion			tions		

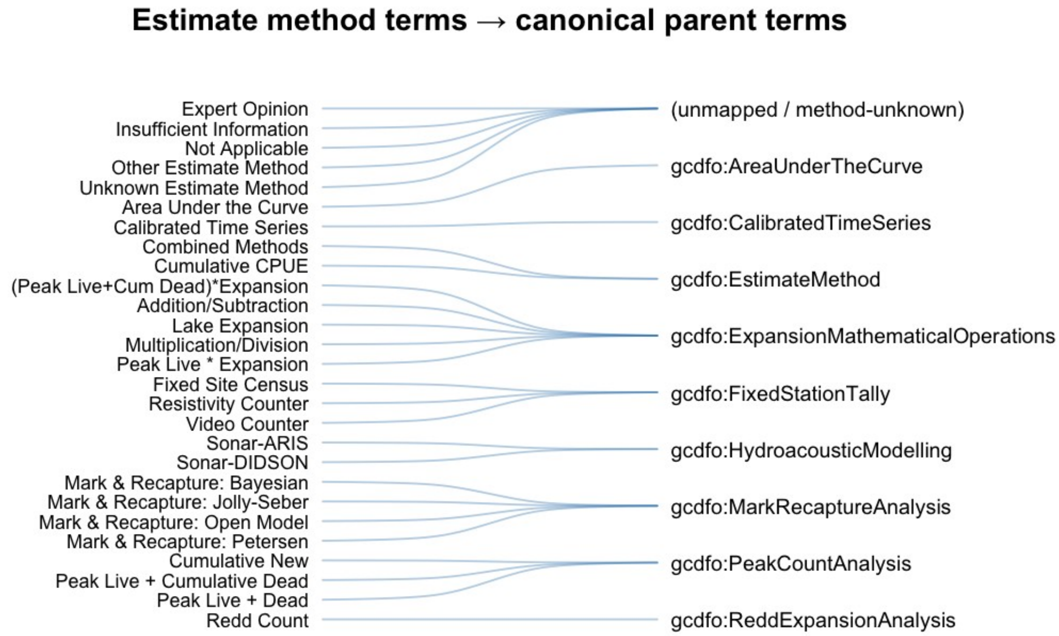

Figure 4. Sankey-style summary of how legacy NuSEDS estimate-method terms converge to canonical parent terms used by the updated guidance.

C.3 Type-to-accuracy/precision lookup used in the Escapement Estimate Type Toolkit

The Escapement Estimate Type Toolkit includes a compact lookup that can be used when populating legacy-style accuracy/precision class fields alongside estimate Type. This is a compatibility aid for workflows that still carry those legacy labels; it does **not** change the Type assignment itself.

Table 10: Escapement Estimate Type Toolkit lookup from estimate Type to legacy-style accuracy/precision class labels, included as a compatibility aid for workflows that still populate those fields.

Estimate Type	accuracy_class	precision_class
1	actual_very_high	plus_minus_zero
2	actual_or_assigned_high	high_to_moderate
3	assigned_range_med_high	assigned_med_high
4	unknown_assumed_constant	unknown_assumed_constant
5	unknown_assumed_highly_variable	unknown_assumed_highly_variable
6	medium_to_high	unknown

Appendix D: NuSEDS method-term summary provenance (SEN exploratory)

This appendix documents the exploratory method-vocabulary evidence used to build the crosswalk logic in Appendix C and to interpret the frequency snapshot shown in Figure 2 of the main text.

D.1 Source and processing notes

The method-term review is intended as implementation context, not as a frozen inventory of future NuSEDS values. The same source and processing notes applied to the Figure 2 frequency summary and to the Appendix C crosswalk review.

Table 11: Source and processing notes for the exploratory NuSEDS method-term review used in Figure 2 and Appendix C.

Item	Details
Source extract	All-area SEN method extract assembled for this report.
Fields reviewed	ENUMERATION_METHODS and ESTIMATE_METHOD.
Year filter	Records with ANALYSIS_YR between 1990 and 2025.
Term handling	Multi-valued cells split on semicolons, commas, and slashes; whitespace normalized before counting.
Primary outputs	Figure 2 in the main text (top terms) and the interpretive crosswalk context summarized in Appendix C.

D.2 Interpretation note

Because these summaries are derived from an exploratory extract rather than from a versioned public release, they should be interpreted as a dated snapshot that helps explain the crosswalk design—not as a guarantee that the same term frequencies will persist as NuSEDS continues to evolve.

DFO Pacific Science. 2026. Metasalmon: Tools for salmon ontology and NuSEDS method workflows. R package.English, K.K., Bocking, R.C., and Irvine, J.R. 1992. A robust procedure for estimating salmon escapement based on the area-under-the-curve method. Canadian Journal of Fisheries and Aquatic Sciences 49(10): 1982–1989.Fisheries and Oceans Canada. 2005. Canada’s policy for conservation of wild pacific salmon. Fisheries; Oceans Canada.Fisheries and Oceans Canada. 2016. NuSEDS–new salmon escapement database system. Open Government Portal dataset.Fisheries and Oceans Canada. 2024. Rapid status approximations for pacific salmon derived from integrated expert assessments under fisheries and oceans canada wild salmon policy. Canadian science advisory secretariat science response 2024/004. Fisheries; Oceans Canada.Fisheries and Oceans Canada. 2025. NuSEDS (new salmon escapement database system) data dictionary (english).Hilborn, R., Bue, B.G., and Sharr, S. 1999. Estimating spawning escapements from periodic counts: A comparison of methods. Canadian Journal of Fisheries and Aquatic Sciences 56(5): 888–896.Hill, R.A. 1997. Optimizing aerial count frequency for the area-under-the-curve method of estimating escapement. North American Journal of Fisheries Management 17(2): 461–466.Holmes, J.A., Cronkite, G., and Enzenhofer, H.J. 2005. Feasibility of deploying a dual-frequency identification sonar (DIDSON) system to estimate salmon spawning ground escapement in major tributary systems of the fraser river, british columbia. Canadian technical report of fisheries and aquatic sciences 2592. Fisheries; Oceans Canada.Holt, K.R., and Cox, S.P. 2008. Evaluation of visual survey methods for monitoring pacific salmon (oncorhynchus spp.) escapement in relation to conservation guidelines. Canadian Journal of Fisheries and Aquatic Sciences 65(2): 212–226.Hyatt, K.D. 1997. Salmon escapement classification key. Internal memo, Fisheries and Oceans Canada, Pacific Biological Station.Jones, E.L., Quinn, T.J., and Van Alen, B.W. 1998. Observer accuracy and precision in aerial and foot survey counts of pink salmon in a southeast alaska stream. North American Journal of Fisheries Management 18(4): 832–846.Korman, J., Ahrens, R.N.M., Higgins, P.S., and Walters, C.J. 2002. Effects of observer efficiency, arrival timing, and survey life on estimates of escapement for steelhead trout (oncorhynchus mykiss) derived from repeat mark-recapture experiments. Canadian Journal of Fisheries and Aquatic Sciences 59(7): 1116–1131.Millar, R.B., McKechnie, S., and Jordan, C.E. 2012. Simple estimators of salmonid escapement and its variance using a new area-under-the-curve method. Canadian Journal of Fisheries and Aquatic Sciences 69(6).Parken, C.K., Bailey, R.E., and Irvine, J.R. 2003. Incorporating uncertainty into area-under-the-curve and peak count salmon escapement estimation. North American Journal of Fisheries Management 23(1): 78–90.Parsons, A.L., and Skalski, J.R. 2010. Quantitative assessment of salmonid escapement techniques. Reviews in Fisheries Science 18(4): 301–314.Pestal, G., MacDonald, B.L., Grant, S.C.H., and Holt, C.A. 2023. State of the salmon: Rapid status assessment approach for pacific salmon under canada’s wild salmon policy. Canadian technical report of fisheries and aquatic sciences 3570. Fisheries; Oceans Canada.Schwarz, C.J., Bailey, R.E., Irvine, J.R., and Dalziel, F.C. 1993. Estimating salmon spawning escapement using capture–recapture methods. Canadian Journal of Fisheries and Aquatic Sciences 50(6): 1181–1197.See, K.E., Kinzer, R.N., and Ackerman, M.W. 2021. State-space model to estimate salmon escapement using multiple data sources. North American Journal of Fisheries Management 41(5): 1360–1374.Vélez-Espino, L.A., and others. 2010. Mark-recapture experiment for the 2009 chinook salmon spawning escapement in the atnarko river. Canadian manuscript report of fisheries and aquatic sciences 2930. Fisheries; Oceans Canada.Walters, C.J., and Ludwig, D. 1981. Effects of measurement errors on the assessment of stock–recruitment relationships. Canadian Journal of Fisheries and Aquatic Sciences 38(6): 704–710.